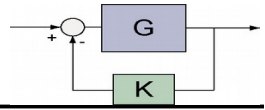


ME 2801

Introduction to Control Systems

W4: Feedback Design Example

Feedback Design Example

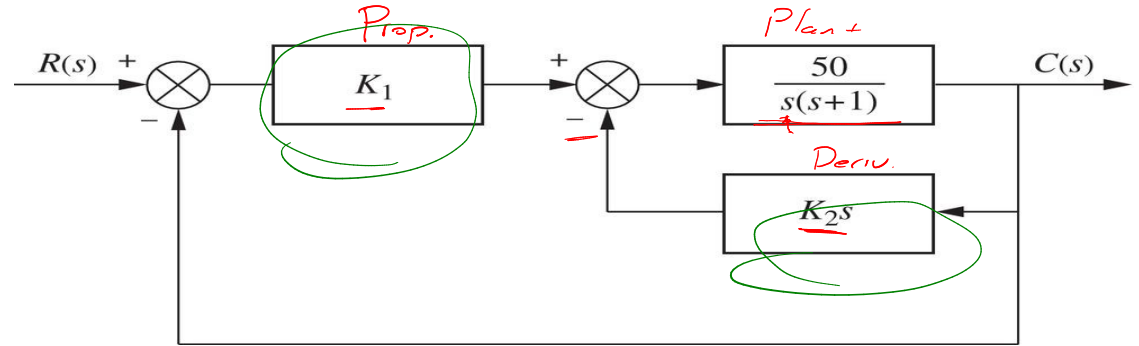


Design Goal:

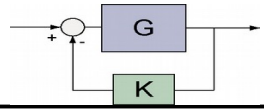
- $T_p = 1s$
- $T_s = 2s$

Control Design:

- Choose gains K_1 and K_2 to satisfy the design constraints

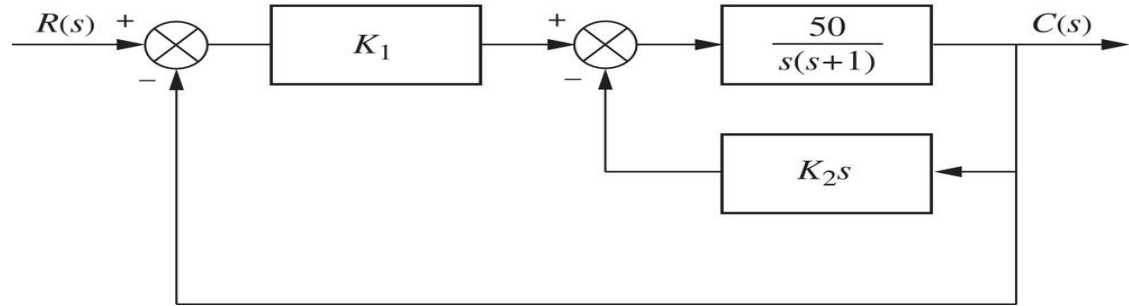


Feedback Design Example



Design Goal:

- $T_p = 1s$
- $T_s = 2s$



Control Design:

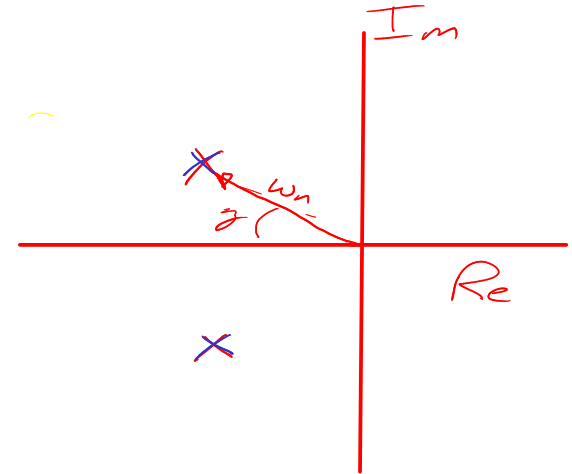
- Choose gains K_1 and K_2

$$T_p = \frac{\pi}{\omega_n \sqrt{(1-\zeta^2)}} = \frac{\pi}{\omega_d}$$

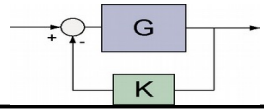
$$T_s = 4 \frac{1}{\zeta \omega_n}$$

$$\zeta = 0.537$$

$$\omega_n = 3.724 \text{ rad/s}$$



Feedback Design Example

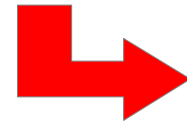
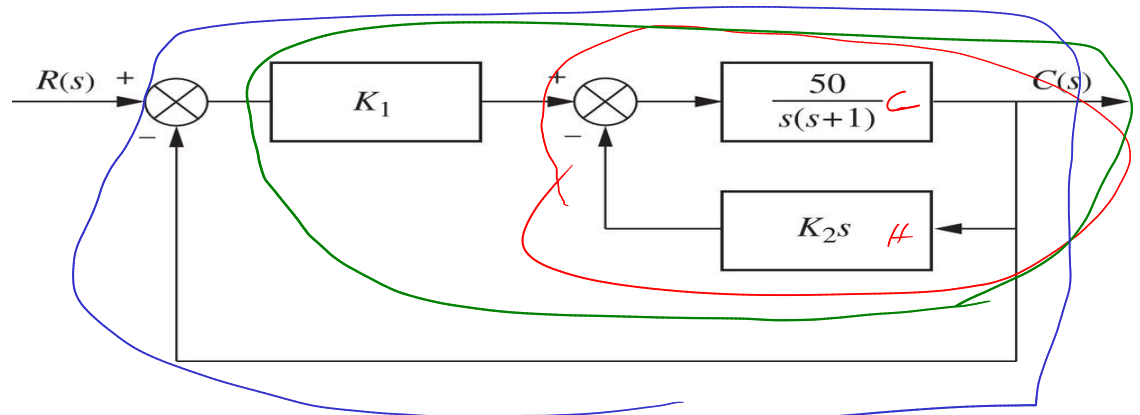


Design Goal:

- $T_p = 1s$
- $T_s = 2s$

Control Design:

- Choose gains K_1 and K_2



$$T(s) = \frac{50(K_1)}{s^2 + (50K_2 + 1)s + 50K_1}$$

Closed-Loop TF
 $s^2 + 2\zeta\omega_n s + \omega_n^2$

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} = \frac{\pi}{\omega_d}$$

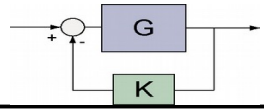
$$T_s = 4 \frac{1}{\zeta \omega_n}$$



$$\zeta = 0.537$$

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Feedback Design Example

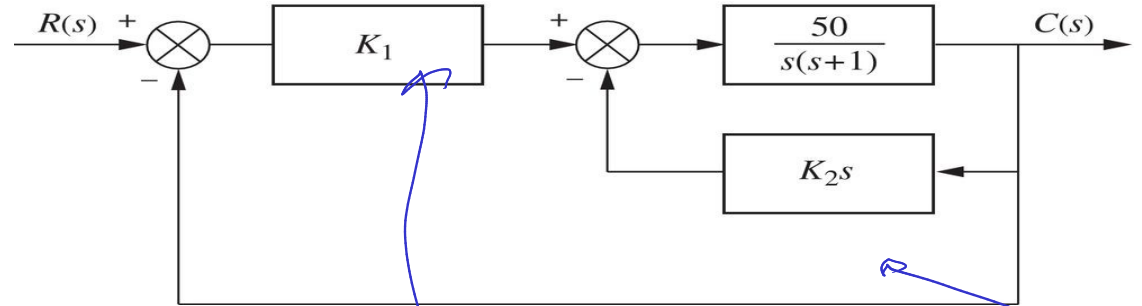


Design Goal:

- $T_p = 1s$
- $T_s = 2s$

Control Design:

- Choose gains K_1 and K_2



$$T_p = \frac{\pi}{\omega_n \sqrt{(1 - \zeta^2)}} = \frac{\pi}{\omega_d}$$

$$T_s = 4 \frac{1}{\zeta \omega_n}$$

$$\zeta = 0.537$$

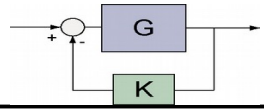
$$\omega_n = 3.724 \text{ rad/s}$$

$$T(s) = \frac{50(K_p)}{s^2 + \underbrace{(50K_d + 1)}_{2\zeta\omega_n} s + \underbrace{50K_p}_{\omega_n^2}}$$

$$K_1 = K_p = 0.277$$

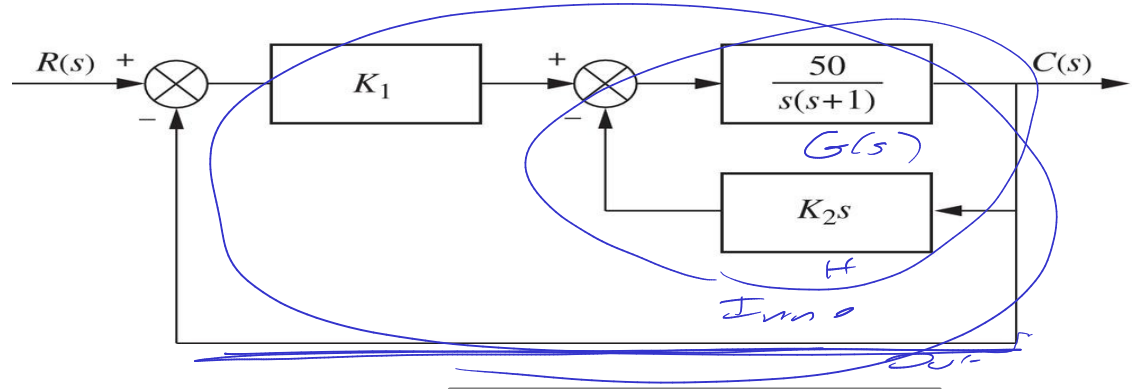
$$K_2 = K_d = 0.06$$

Check With MATLAB



```
K1 = 0.277; % Kp
K2 = 0.06; % Kd

G = tf(50,[1 1 0])
H = K2*tf([1 0],1)
```



$$K_p = K_1 = 0.277$$

$$K_d = K_2 = 0.06$$



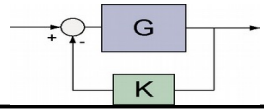
$$\zeta = 0.537$$

$$\omega_n = 3.724 \text{ rad/s}$$

$$T_p = 1.0 \text{ s}$$

$$T_s = 2.0 \text{ s}$$

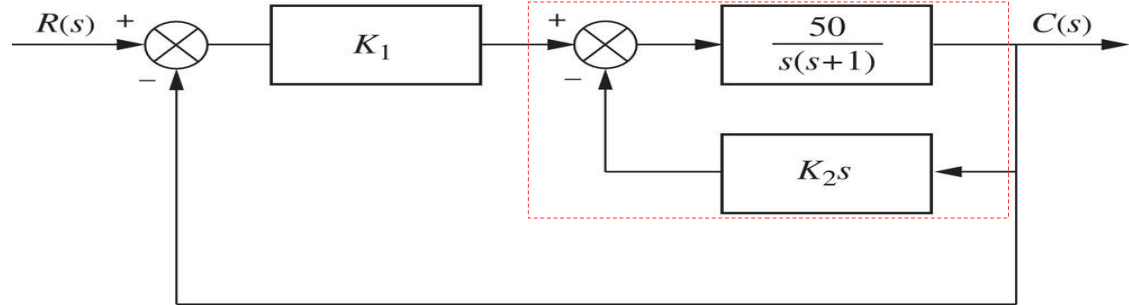
Check With MATLAB



```
K1 = 0.277; % Kp
K2 = 0.06; % Kd
```

```
G = tf(50,[1 1 0])
H = K2*tf([1 0],1)
```

```
% Inner Loop
Ginner = feedback(G,H)
```



$$K_p = K_1 = 0.277$$

$$K_d = K_2 = 0.06$$



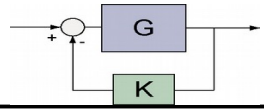
$$\zeta = 0.537$$

$$\omega_n = 3.724 \text{ rad/s}$$

$$T_p = 1.0 \text{ s}$$

$$T_s = 2.0 \text{ s}$$

Check With MATLAB

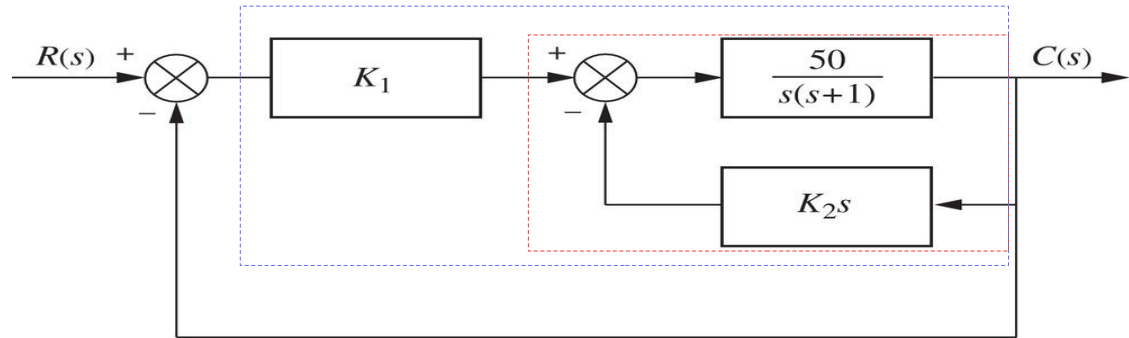


```
K1 = 0.277; % Kp
K2 = 0.06; % Kd
```

```
G = tf(50,[1 1 0])
H = K2*tf([1 0],1)
```

```
% Inner Loop
Ginner = feedback(G,H)
```

```
% Cascade
Ginner2 = K1*Ginner;
```



$$K_p = K_1 = 0.277$$

$$K_d = K_2 = 0.06$$



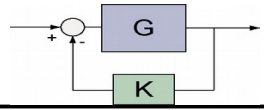
$$\zeta = 0.537$$

$$\omega_n = 3.724 \text{ rad/s}$$

$$T_p = 1.0 \text{ s}$$

$$T_s = 2.0 \text{ s}$$

Check With MATLAB



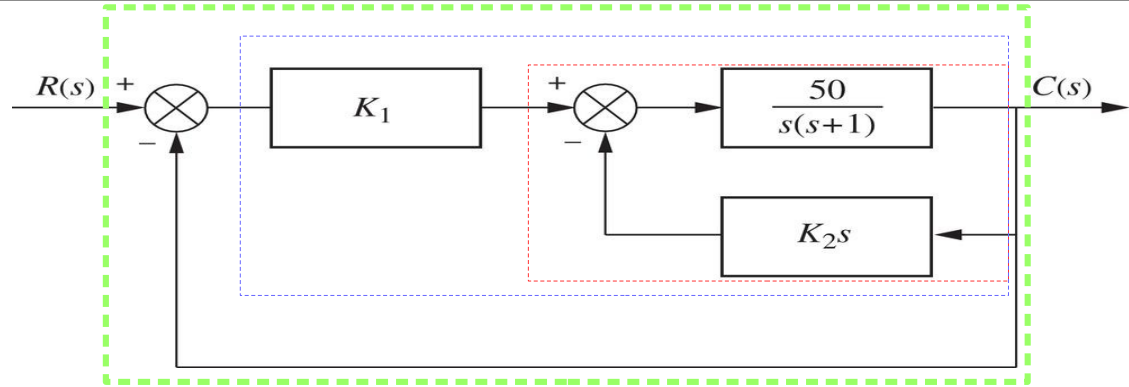
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K1 = 0.277; % Kp
K2 = 0.06; % Kd
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G = tf(50,[1 1 0])
H = K2*tf([1 0],1)
```

```
% Inner Loop
Ginner = feedback(G,H)
```

```
% Cascade
Ginner2 = K1*Ginner;
```

```
% Outer Loop
Gclosed = feedback(Ginner2,1)
```



$$K_p = K_1 = 0.277$$

$$K_d = K_2 = 0.06$$



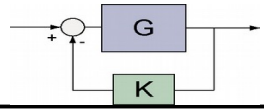
$$\zeta = 0.537$$

$$\omega_n = 3.724 \text{ rad/s}$$

$$T_p = 1.0 \text{ s}$$

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Check With MATLAB



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K1 = 0.277; % Kp
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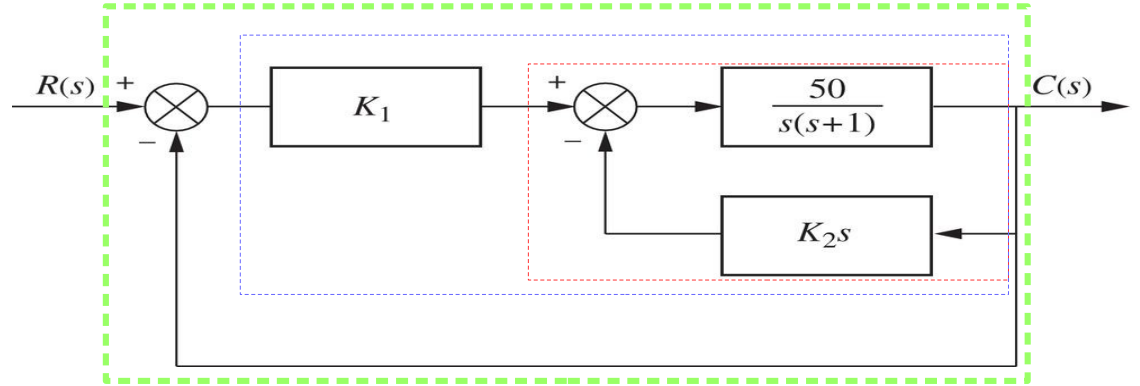
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G = tf(50,[1 1 0])
H = K2*tf([1 0],1)
```

```
% Inner Loop
Ginner = feedback(G,H)
```

```
% Cascade
Ginner2 = K1*Ginner;
```

```
% Outer Loop
Gclosed = feedback(Ginner2,1)
```

```
>> stepinfo(Gclosed)
RiseTime: 0.4607
SettlingTime: 1.5581
SettlingMin: 0.9082
SettlingMax: 1.1350
Overshoot: 13.4954
Undershoot: 0
Peak: 1.1350
PeakTime: 0.9901
```



$$K_p = K_1 = 0.277$$

$$K_d = K_2 = 0.06$$



$$\zeta = 0.537$$

$$\omega_n = 3.724 \text{ rad/s}$$

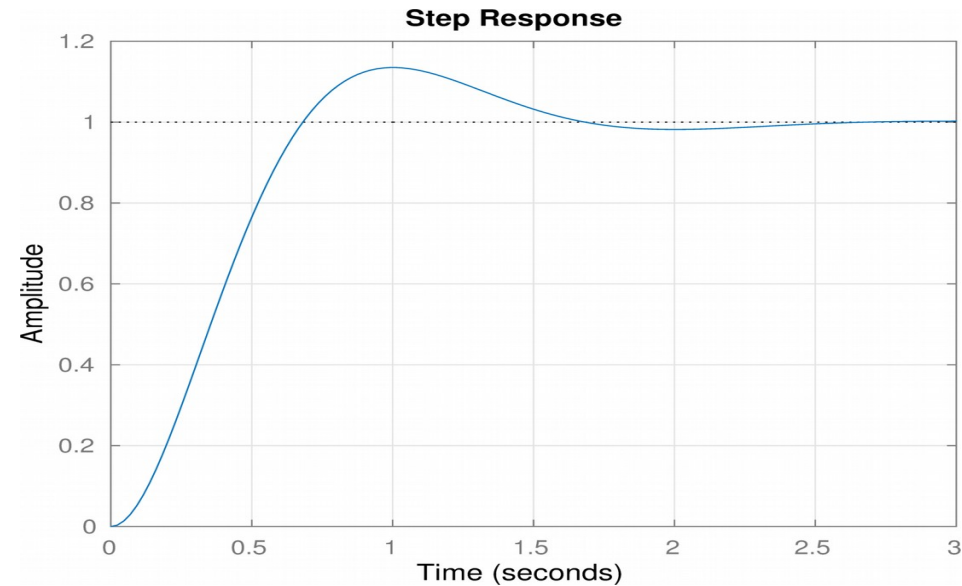
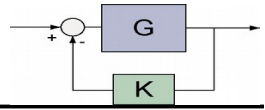
$$T_p = 1.0 \text{ s}$$

$$T_s = 2.0 \text{ s}$$

why?

Peak: 1.1350
PeakTime: 0.9901

Check With MATLAB



```
>> stepinfo(Gclosed)
ans =
    RiseTime: 0.4607
    SettlingTime: 1.5581
    SettlingMin: 0.9082
    SettlingMax: 1.1350
    Overshoot: 13.4954
    Undershoot: 0
    Peak: 1.1350
    PeakTime: 0.9901
```

why?

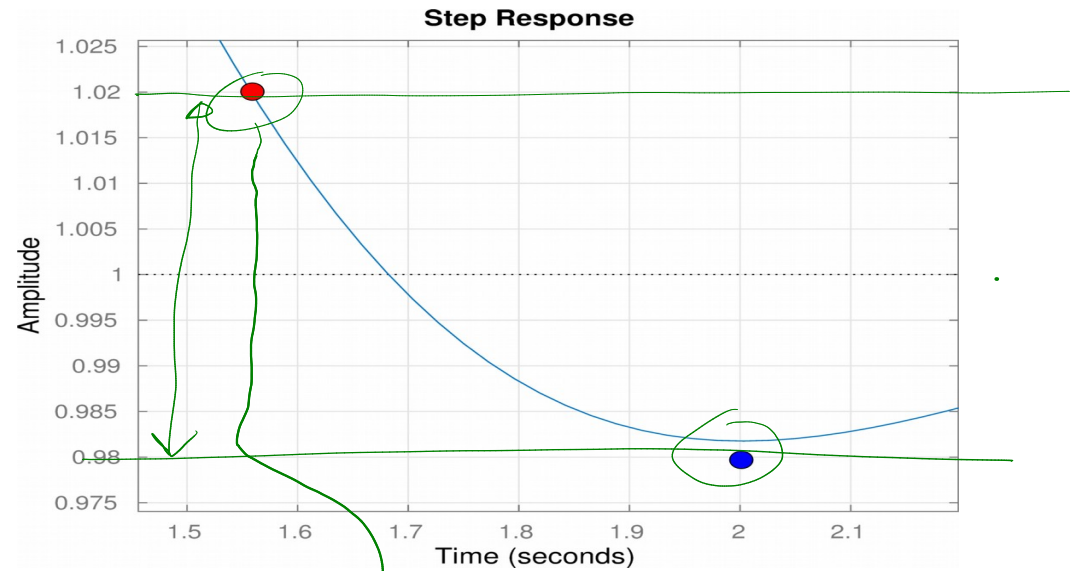
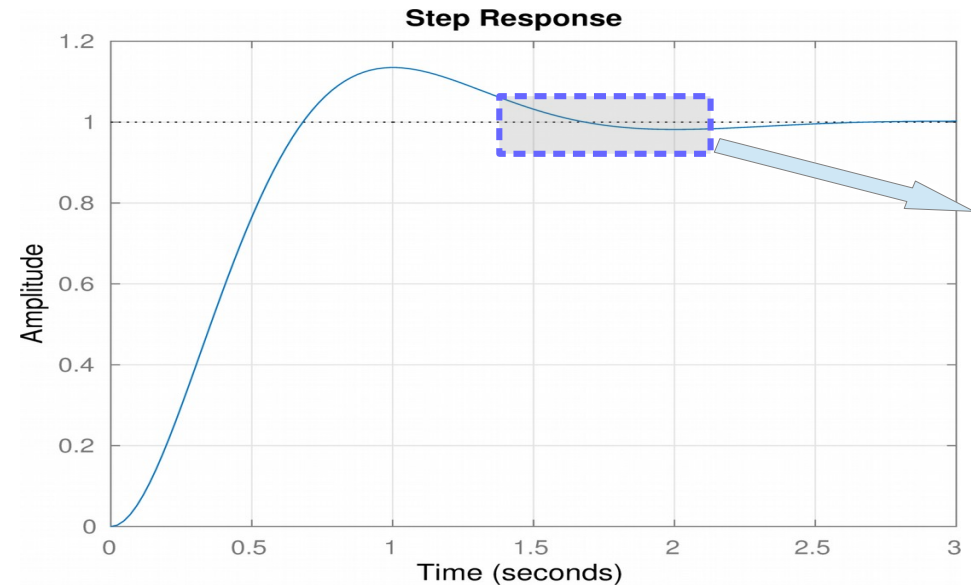
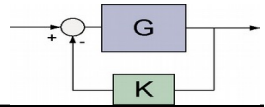
$$\zeta = 0.537$$

$$\omega_n = 3.724 \text{ rad/s}$$

$$T_p = 1.0 \text{ s}$$

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Check With MATLAB



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>> stepinfo(Gclosed)
ans =
    RiseTime: 0.4607
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    Undershoot: 0
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```

why?

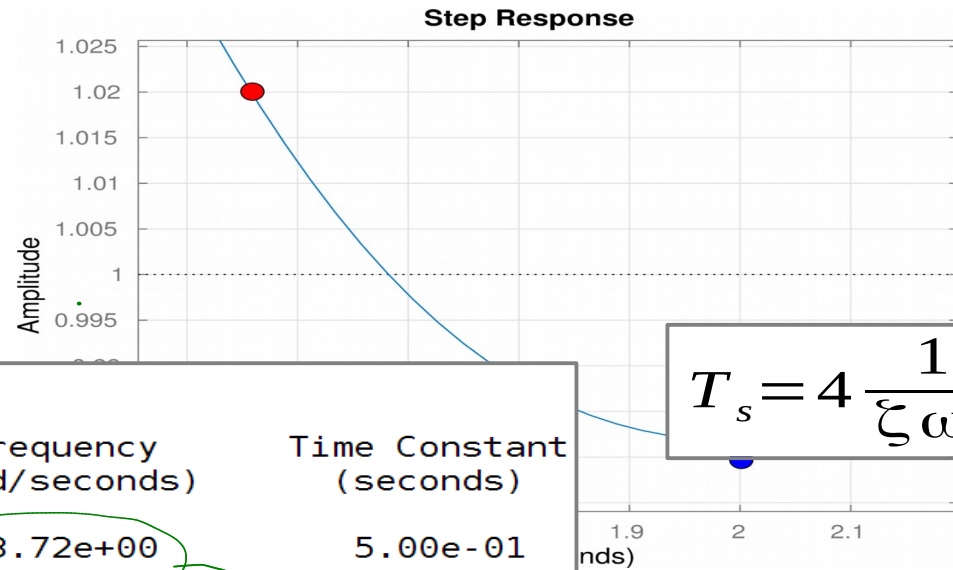
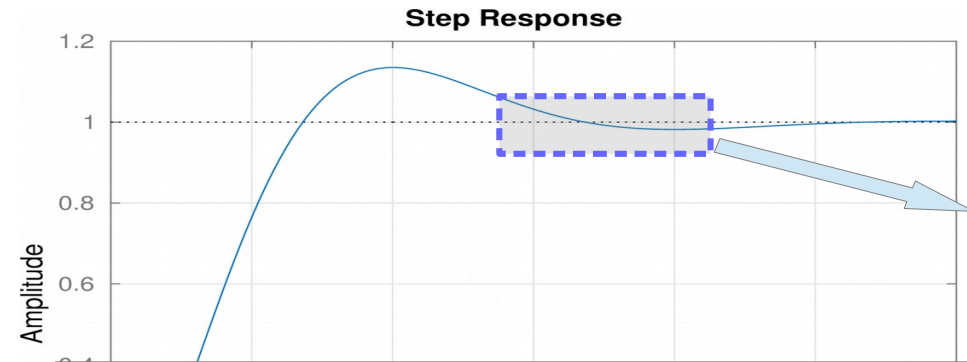
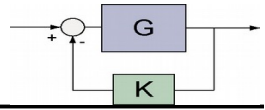
$$\zeta = 0.537$$

$$\omega_n = 3.724 \text{ rad/s}$$

$$T_p = 1.0 \text{ s}$$

$$T_s = 2.0 \text{ s}$$

Check With MATLAB



```
>> damp(Gclosed)
```

Pole	Damping	Frequency (rad/seconds)	Time Constant (seconds)
-2.00e+00 + 3.14e+00i	5.37e-01	3.72e+00	5.00e-01
-2.00e+00 - 3.14e+00i	5.37e-01	3.72e+00	5.00e-01

$$T_s = 4 \frac{1}{\zeta \omega_n}$$

```
>> stepinfo(Gclosed)
ans =
```

```

    RiseTime: 0.4607
    SettlingTime: 1.5581
    SettlingMin: 0.9082
    SettlingMax: 1.1350
    Overshoot: 13.4954
    Undershoot: 0
    Peak: 1.1350
    PeakTime: 0.9901
```

why?

$$\zeta = 0.537$$

$$\omega_n = 3.724 \text{ rad/s}$$

$$T_p = 1.0 \text{ s}$$

$$T_s = 2.0 \text{ s}$$